- ELEMENTI FINITI

RIASSUNTONE 1D

TROVA $u \in V$

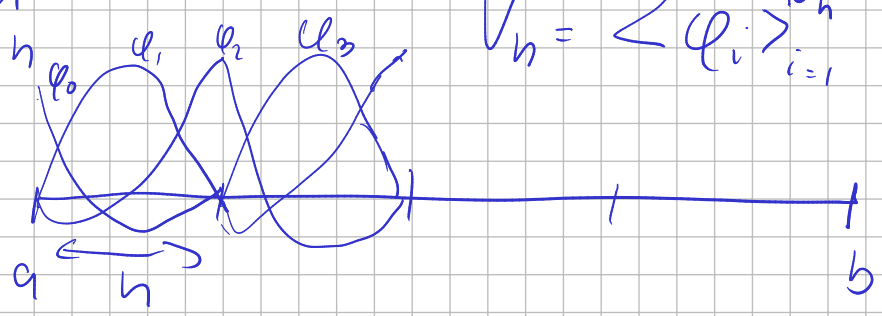
$$-u'' = f \quad \rightsquigarrow \quad \int u' v' = \int f v \qquad \forall v \in V$$

<u>DISCRETA</u> $V_h \subset V$

$$\int u_h' v_h' = \int f \cdot v_h \qquad \forall v_h \in V_h$$

TROVA $u_h \in V_h$

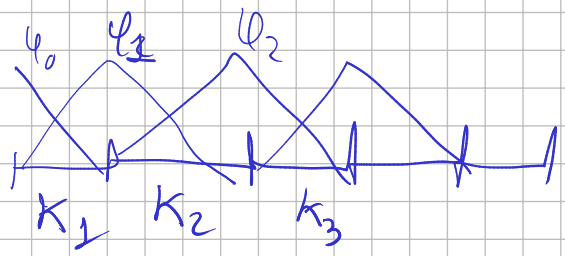
$$V_h = X_h^p \qquad V_h = \langle \varphi_i \rangle_{i=1}^{N_h}$$



$$u_h(x) = \sum_{j=1}^{N_h} u_j \varphi_j(x) \qquad \int u_h' \cdot \varphi_i' = \int f \cdot \varphi_i \quad \forall i = 1, \dots, N_h$$

$$\sum_{j=1}^{N_h} \underbrace{\int \varphi_j' \varphi_i' \, dx}_{A_{ij}} u_j = \underbrace{\int f \varphi_i \, dx}_{f_i}$$

$$\boxed{\underline{A}\, \underline{\varphi} = \underline{f}}$$

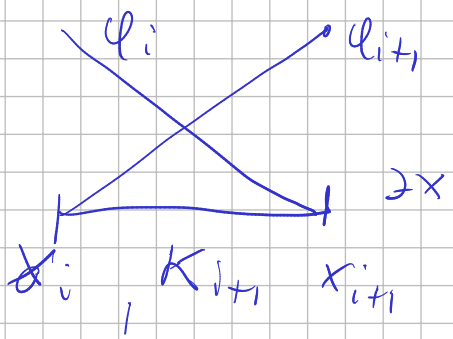


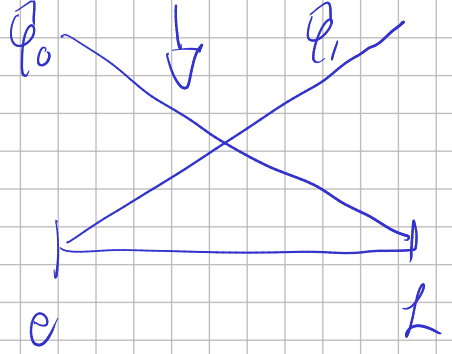
ELEMENTO DI RIFERIMENTO $[0,1]$

PER CALCOLARE GLI INTEGRALI

$$T_i : K_i \to [0,1]$$

$$T_i^{-1} : [0,1] \to K_i$$





$$\int \hat{\varphi}_0' \, \hat{\varphi}_1' = \int_0^1 x'(1-x)' = \int_0^1 1(-1) = -1$$

$\Rightarrow$

$$\int_{x_i}^{x_{i+1}} \varphi_i' \, \varphi_{i+1}' \, dx = \int_0^1 \underbrace{\frac{\partial \xi}{\partial x}}_{\frac{1}{\Delta x}} \partial_\xi \hat{\varphi}_0 \cdot \underbrace{\frac{\partial \xi}{\partial x}}_{\frac{1}{\Delta x}} \partial_\xi \hat{\varphi}_1 \, \underbrace{\frac{dx}{d\xi}}_{\Delta x} \, d\xi$$

$\partial_x \varphi_i$

$$= \frac{1}{\Delta x} \underbrace{(-1)}$$

$\Rightarrow$ A SBARRA, FACILE DA CALCOLARE.

$\boxed{2D}$ PROBLEMA ELLITTICO TIPO

$$-\Delta u = f \qquad \underline{x} = (x,y)$$

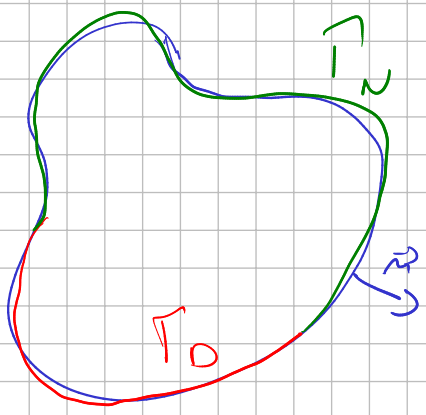
$$-\partial_{xx} u(x,y) - \partial_{yy} u(x,y) = f(x,y) \qquad (x,y) \in \Omega \subset \mathbb{R}^2$$

$$\int_\Omega \varphi(-\Delta u) \, d\underline{x} = \int_\Omega f(\underline{x}) \cdot \varphi(\underline{x}) \, dx$$

1D INTEGRARE PER PARTI

2D TEOREMA DIVERGENZA

$\varphi \in V$



$$-\int_{\partial\Omega} \varphi \cdot \underbrace{\nabla u \cdot \vec{n}}_{} \, d\Gamma + \int_\Omega \nabla\varphi \cdot \nabla u \, dx = \int f \cdot \varphi \, dx$$

$\Rightarrow$ FORMA DEBOLE (+ BC)

BC: DI DIRICHLET $u(x) = u_D(x) \quad \forall x \in \Gamma_D$

$\rightarrow \nabla u \cdot \vec{n} = g_N(x) \quad \forall x \in \Gamma_N$

$\rightarrow \varphi \in V = H^1(\Omega) \cap \{ \varphi(x) : \varphi(x) = 0 \ \forall x \in \Gamma_D \}$

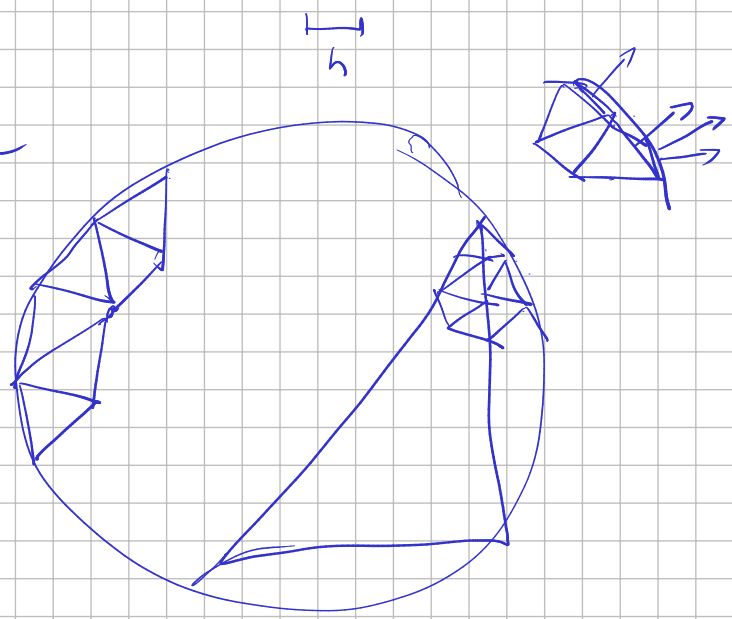
$u \in V_D = H^1(\Omega) \cap \{ u(x) : u(x) = u_D(x) \ \forall x \in \Gamma_D \}$

$$-\int_{\Gamma_N} \varphi(x)\, g_N(x)\, d\Gamma + \int_\Omega \nabla\varphi \cdot \nabla u \, dx = \int_\Omega f \varphi \, dx$$

ELEMENTI FINITI 2D

$\Omega_h \approx \Omega$

$$\Omega_h = \left(\bigcup_{K \in \mathcal{T}_h} \overset{\circ}{K} \right)$$



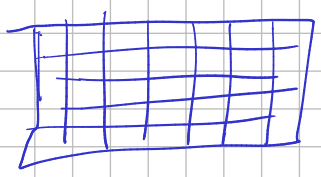
$\mathcal{T}_h$ TRIANGOLAZIONE (INSIEME DI TUTTI GLI ELEMENTI K)

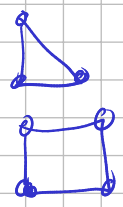
$K_1 \cap K_2 = \emptyset \qquad K_1 \neq K_2 \qquad \forall K_1, K_2 \in \mathcal{T}_h$

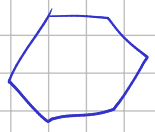
$h_K := \max_{x,y \in K} \|x - y\|_2 =: \mathrm{diam}(K)$

$h = \max_K h_K$

- TRIANGOLI
- QUADRILATERI
- POLIGONI VARI

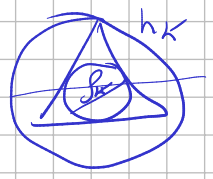


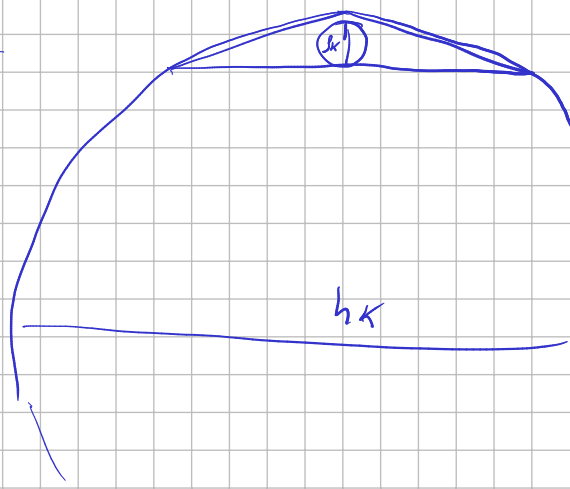




ρ_K = DIAMETRO CERCHIO INSCRITTO NEI TRIANGOLI

$$\boxed{\frac{h_K}{\rho_K}} \leq \delta$$

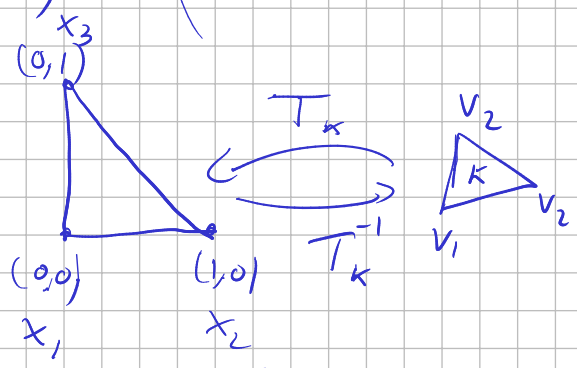




DELAUNAY ⇒ TRIANGOLARIZZAZIONE

TRIANGOLO (DI Ù SEMPLICE)

ELEMENTO DI RIFERIMENTO



$$\mathbb{P}^p = \left\{ f(x,y) = \sum_{i,j:\ i+j \leq p} a_{ij}\, x^i y^j \ :\ a_{ij} \in \mathbb{R} \right\}$$

$$\mathbb{P}^1 = \{ ax + by + c \}$$

LINEARI IN 2D

$$\begin{pmatrix} 0,0 \\ 1,0 \\ 0,1 \end{pmatrix}$$

$$\dim(\mathbb{P}^1) = 3$$

$$\mathbb{P}^2 = \{ a_{00} + a_{10}x + a_{01}y + a_{11}xy + a_{20}x^2 + a_{02}y^2 \}$$

$$\mathbb{Q}^p = \left\{ f(x,y) = \sum_{\substack{i,j: \\ \max(i,j) \leq p}} a_{ij}\, x^i y^j \ :\ a_{ij} \in \mathbb{R} \right\}$$

$$\mathbb{Q}^1 = \{ a_{00} + a_{10}x + a_{01}y + a_{11}xy \}$$

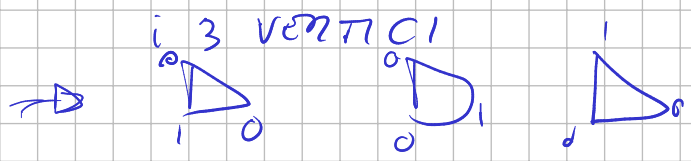
BILINEARI

$$\dim(\mathbb{Q}^1) = 4$$

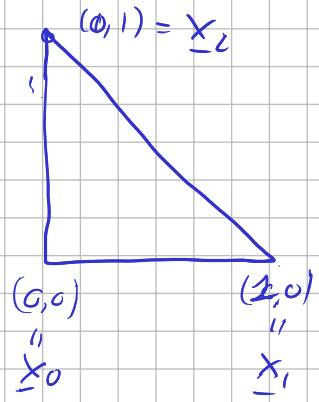
GOAL

TRIANGOLO ⇒ $\mathbb{P}^p$

es. $\mathbb{P}^1$ 3 FUNZIONI DI BASE LAGRANGIANE INTERPOLINO I 3 VERTICI



$\mathbb{P}^1$ TRIANGOLO DI RIFERIMENTO



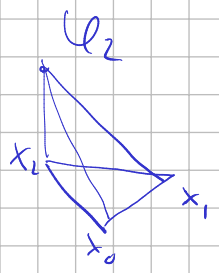
$$\hat{\varphi}_i(\underline{x}_j) = \delta_{ij} = \begin{cases} 1 & i=j \\ 0 & i\neq j \end{cases} \qquad i,j \in \{0,1,2\}$$

$$\hat{\varphi}_i(\underline{x}) = a_i x + b_i y + c_i \qquad i=0,1,2$$

$$\begin{cases} \varphi_i(\underline{x}_0) = \delta_{i0} \\ \varphi_i(\underline{x}_1) = \delta_{i1} \\ \varphi_i(\underline{x}_2) = \delta_{i2} \end{cases} \Rightarrow$$

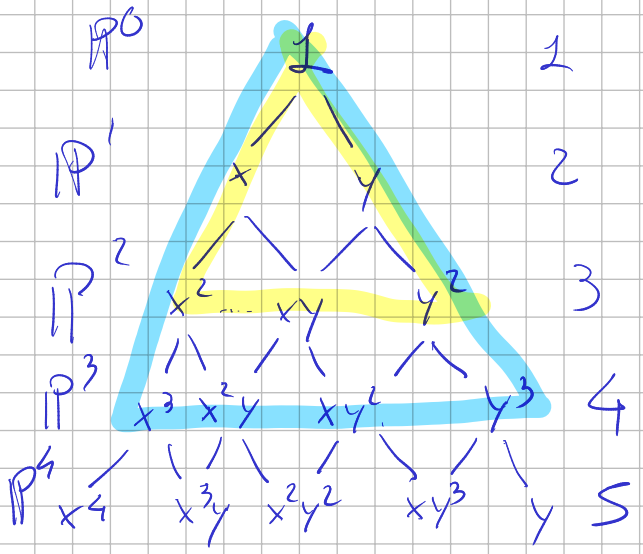
SYS LINEARE
3 INCOGNITE
3 equazioni

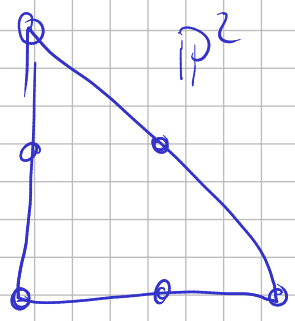
$$\varphi_0(x,y) = 1 - x - y$$
$$\varphi_1(x,y) = x$$
$$\varphi_2(x,y) = y$$

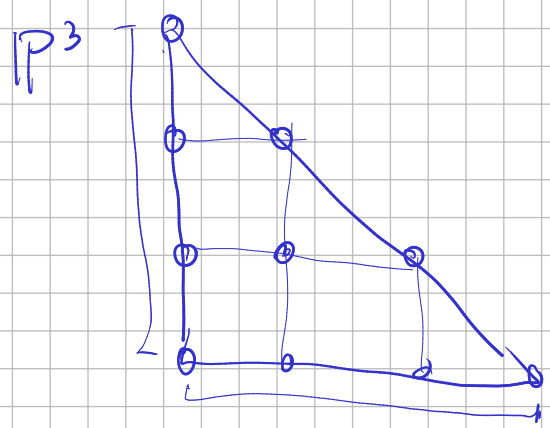


$\boxed{\mathbb{P}^2}$ $\quad 1, x, y, x^2, xy, y^2 \quad 6 \qquad \dim(\mathbb{P}^2) = 6$

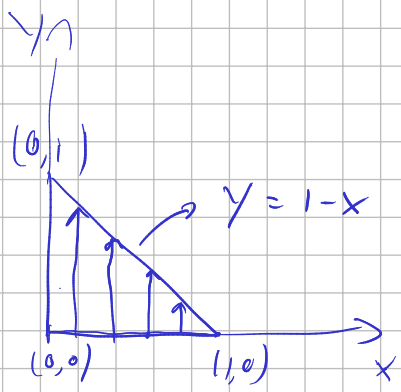
$$\dim(\mathbb{P}^p) = \sum_{k=1}^{p+1} k = \frac{(p+1)(p+2)}{2}$$



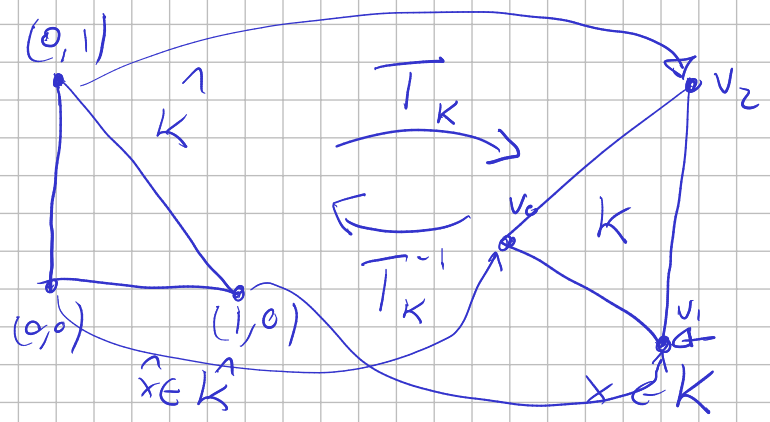




$\hat{K} =$



$$\int_{\hat{K}} \nabla \hat{\varphi}_i \cdot \nabla \hat{\varphi}_j = \int_0^1 \int_0^{1-x} \nabla \hat{\varphi}_i(x) \cdot \nabla \hat{\varphi}_j(x) \, dy \, dx$$



$$x = T_K(\hat{x}) = \underset{\mathbb{R}^{2\times 2}}{A}\,\hat{x} + \underset{\mathbb{R}^2}{b}$$

$$\underbrace{4}_{2\times 2} + \underbrace{2}_{2} = 6$$

6 INCOGNITE : $A, b \in \mathbb{R}^{2\times 2} \times \mathbb{R}^2$

6 EQUAZIONI : 2 x OGNI VERTICE V_0, V_1, V_2 (x e y)

- $\exists!$ SOLUZ. SYS LINEARE se V_0, V_1, V_2 NON SONO ALLINEATI
- ESISTONO FORMULE ESPLICITE

$$\frac{\partial}{\partial x} = \frac{\partial \xi}{\partial x} \cdot \frac{\partial}{\partial \xi}$$

$$\hat{x} = T_K^{-1}(x) = A^{-1}(x - b) = A^{-1}x - A^{-1}b$$

$$\int_K \nabla_x \varphi_i(x) \cdot \nabla_x \varphi_j(x) \, dx =$$

$$\det(JT_K) = \det(A)$$

$$\int_{\hat{K}} \underbrace{\nabla_x T_K^{-1}(T_K(\hat{x}))}_{\in \mathbb{R}^2} \cdot \nabla_{\hat{x}} \hat{\varphi}_i(\hat{x}) \cdot \nabla_x T_K^{-1}(T_K(\hat{x})) \cdot \nabla_{\hat{x}} \hat{\varphi}_j(\hat{x}) \det(A) \, d\hat{x}$$

$$\boxed{JT_K = \nabla_{\hat{x}} T_K}$$

$\frac{1}{\Delta x}$ $\frac{1}{\Delta x}$ Δx

$$\int_{\hat{K}} \underbrace{\overbrace{A^{-1}}\ \overbrace{\nabla_{\hat{x}} \hat{\varphi}_i(\hat{x})}^{\in \mathbb{R}^2}}_{} \cdot \underbrace{\overbrace{A^{-1}}\ \overbrace{\nabla_{\hat{x}} \hat{\varphi}_j(\hat{x})}^{\in \mathbb{R}^2}}_{} \underbrace{\det(A)}_{} \, d\hat{x} \rightsquigarrow a_{ij}$$

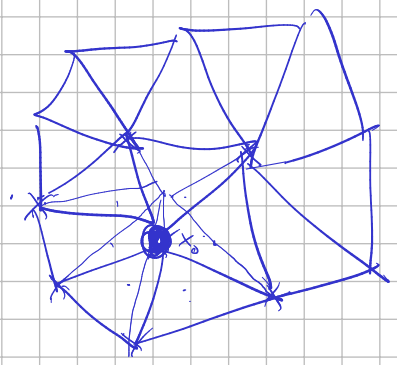
COSTANTI ! $\in \mathbb{R}$ $\in \mathbb{R}$

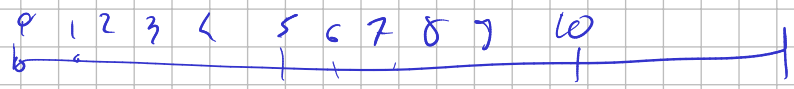
$$\left\{ \begin{array}{l} \int \partial_x \hat{\varphi}_i \, \partial_x \hat{\varphi}_j \\ \int \partial_y \hat{\varphi}_i \, \partial_y \hat{\varphi}_j \\ \int \partial_y \hat{\varphi}_i \, \partial_x \hat{\varphi}_j \\ \int \partial_y \hat{\varphi}_i \, \partial_y \hat{\varphi}_j \end{array} \right\}$$

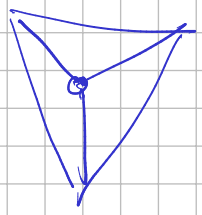
GLI UNICI 4 INTEGRALI ($\forall$ COPPIA DI FUNZIONE DI BASE) CHE DOBBIAMO CALCOLARE

POI MOLTIPLICHIAMO PER DELLE COSTANTI $(A^{-1}, \det(A))$ E SOMMIAMO.

MESH = TRIANGOLAZIONE

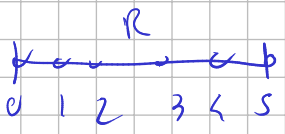


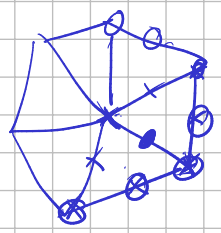


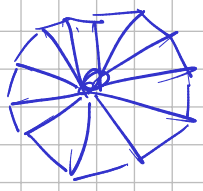


$\mathbb{P}^1$

$\mathbb{P}^2$







STIME ERRORE

$$u \in H^{p+1}(\Omega)$$

$$\|u - u_h\|_{H^1(\Omega)} \leq \frac{M}{\alpha} C h^p \cdot |u|_{H^{p+1}(\Omega)}$$

M è LA COSTANTE DI CONTINUITÀ DELLA FORMA BILIN. a

α è " " " COERCIVITÀ " " " a

C dipende da geometria, δ, NON DA h DIRETTAMENTE

$$\boxed{\frac{h_K}{\rho_K} < \delta}$$

$$\|u - u_h\|_{H^0} = \|u - u_h\|_{L^2} \leq \frac{M}{\alpha} C h^{p+1} |u|_{H^{p+1}(\Omega)}$$